

ARE100B Week 2 Discussion problems

Question 1

Johnson and Johnson are aiming to create a brand new ozempic-like drug that they think will take the world by storm. The cost of developing this drug is \$150 million.

The total cost of supplying the drug, once it's developed is $TC = Q + Q^2$.

Demand for this drug is $Q_D = 20,000 - 0.2P$.

For questions (a) - (d), let J&J operate as a monopolist.

Question 1 (a): Write down J&J's profit function in terms of price

Question 1 (b): Find the optimal price J&J would charge as a monopolist; find the quantity that it will sell.

Question 1 (c): Find the consumer surplus and producer surplus.

Question 1 (d): If J&J can operate as a monopolist, will it develop this drug and bring it to market?

Now assume that if J&J develops the new drug, it would sell it in a competitive market and would earn 10% of total market profits/

Question 1 (e): Find the price and quantity of the drug sold in a competitive market.

Question 1 (f): Find total industry profit (producer surplus) and J&J's profit.

Question 1 (g): Would J&J develop the drug if it new it would have to sell it in a competitive market?

Question 1 (h): If the government could subsidize J&J's drug development, how much would the government have to give J&J so that it would still develop the drug in a competitive market?

Question 2

A monopolist has a marginal cost of 2.

Question 2 (a): Would the monopolist prefer to face consumers with demand $Q_D = 8 - P$ or $Q_D = 7 - \frac{1}{2}P$. In both cases it will sell the same quantity of 3 units.

Question 2 (b): Calculate the elasticities of demand when $Q = 3$ for both demand functions. Use this to justify your answer to part (a).

Question 3

If you know that a monopolist has a marginal cost of \$5, and, at equilibrium, its customers have a price elasticity of -2, find the price the monopolist will charge.

Extra practice problems

Question A

Describe in words, why the monopolist will always charge prices within the elastic portion of demand?

Question B

Let a monopolist's profit function be:

$$\pi = \text{Revenue} - \text{Total cost}$$

Find the first-order/optimal condition of the monopolist setting prices. Describe the optimal condition in terms of an economic trade-off the monopolist needs to make.

Question C

PGE are looking to build a new power grid that is connecting the new techno-optimist city in Solano county. It will cost \$2 million to build.

Once built, the demand for electricity will be $Q_D = 100,000 - P$. PGE has a constant marginal cost selling electricity of \$5.

Question C (a): If PGE can act like a monopolist in terms of selling electricity, what price would it charge?

Question C (b): If PGE can act like a monopolist, would it make the investment to build the new grid?

Question C (c): Suppose CPUC gets wind of PGE trying to charge too much and sets a price PGE has to charge for electricity of \$30. Would PGE still build the new grid at this price?

Question C (d): What is the minimum price that CPUC should set so that PGE still build the new grid?